

RTL Desing and Simulation of a 2x2 Systolic Array for AI Accelerator Using Verilog HDL

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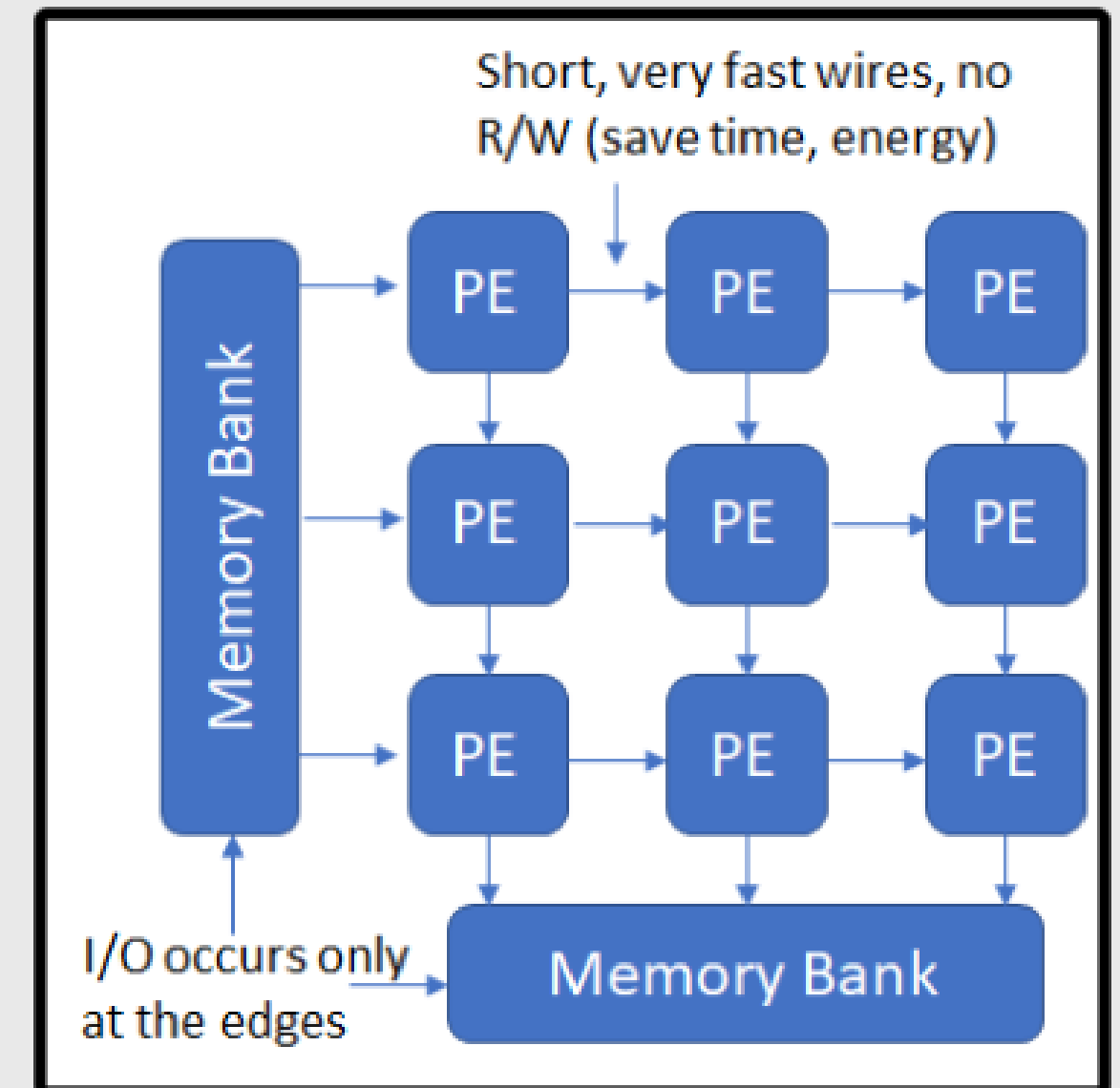
INTRODUCTION

What is a Systolic Array?

- A systolic array is a network of interconnected MAC units.
- Data flows between neighboring processing elements.
- Multiple computations occur simultaneously.
- It supports efficient matrix multiplication.

Why is it Important?

- Enables parallel processing.
- Reduces computation time.
- Improves hardware utilization.
- Widely used in AI accelerators and TPUs (tensor processing unit).



ROLE OF SYSTOLIC

Why is a Systolic Array Important to AI Models?

- AI models mainly perform:
 $C = A \times B$
- A single MAC unit performs one multiplication-accumulation at a time.
- A systolic array contains many MAC units working together in parallel.

Face Recognition Example:

- Face image is converted into numerical matrices.
- AI model performs many matrix multiplications.
- A single MAC unit would process them one by one.
- A systolic array contains many MAC units working simultaneously.
- Multiple matrix calculations happen in parallel.

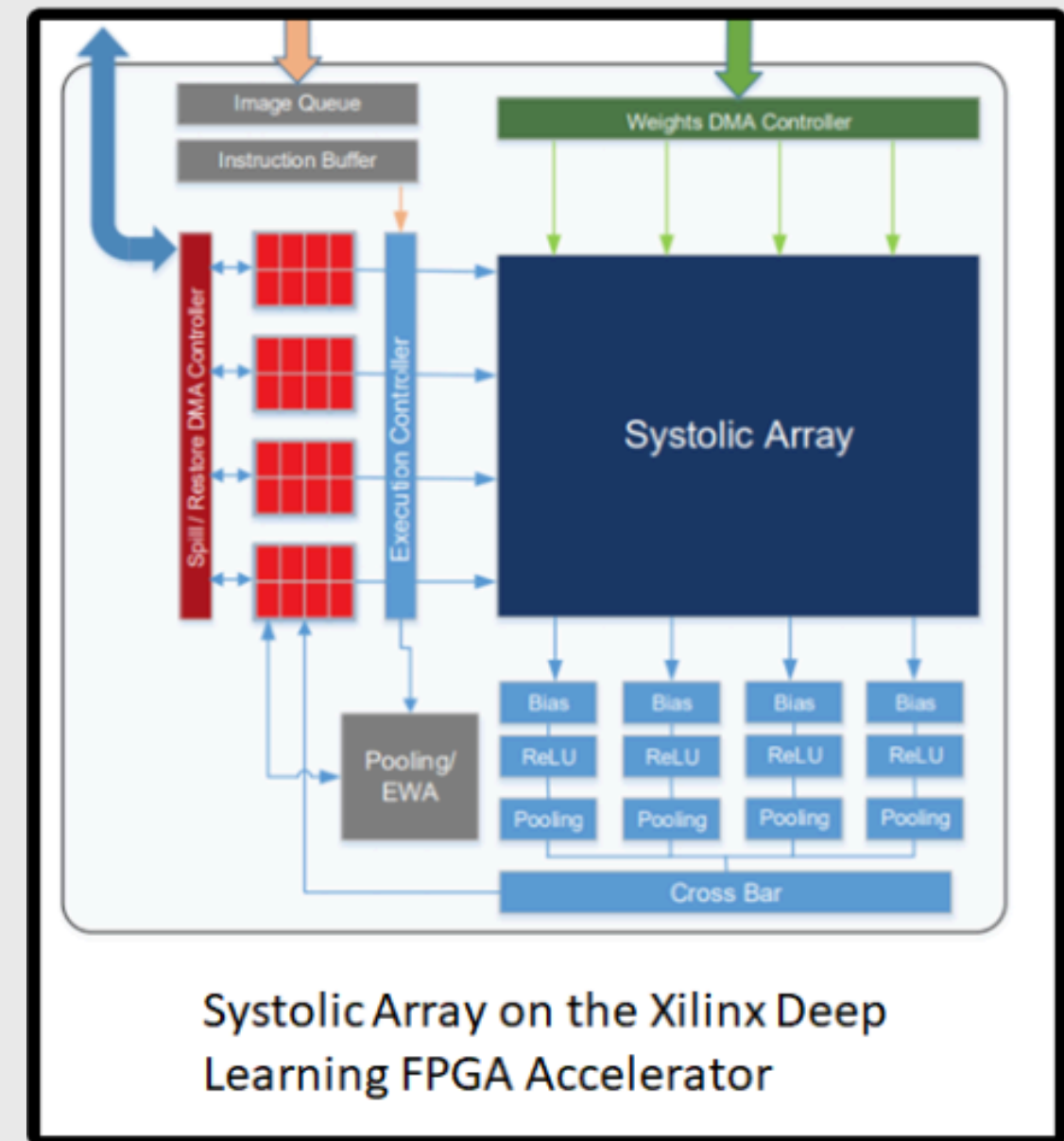
ARCHITECTURE

Data Flow

- Input data A moves from Left → Right.
- Input data B moves from Top → Bottom.
- Partial sums move between neighboring MAC units.
- Data flows rhythmically through the array.

Working Principle

- Data enters the systolic array.
- Each MAC unit performs multiplication and accumulation.
- Results are passed to adjacent MAC units.
- Multiple computations occur simultaneously.
- Final outputs are generated after several clock cycles.



VERILOG SIMULATION

MAC Unit

- Performs Multiply-Accumulate (MAC) operation.
- Multiplies inputs A and B.
- Adds the result to sum_in.
- Stores output in sum_out.
- Reset clears the output.

Systolic Array

- Consists of 4 MAC units.
- MAC units operate simultaneously.
- Demonstrates parallel processing.
- Produces outputs C1, C2, C3, and C4.

The top screenshot shows the Verilog code for a MAC unit and a 2x2 systolic array. The MAC unit code defines inputs for clock, reset, A, B, sum_in, and sum_out, and implements a multiply-accumulate operation. The systolic array code instantiates four MAC units and connects their inputs and outputs. The bottom screenshot shows the same code with simulation results. The output window displays the values of C1, C2, C3, and C4 at different time points, confirming the correct operation of the systolic array.

```
1 module mac_unit (
2     input clk,
3     input reset,
4     input [7:0] A,
5     input [7:0] B,
6     input [15:0] sum_in,
7     output reg [15:0] sum_out
8 );
9
10 always @(posedge clk or posedge reset)
11 begin
12     if(reset)
13         sum_out <= 16'd0;
14     else
15         sum_out <= sum_in + (A * B);
16 end
17
18 endmodule
19
20 // 2x2 SYSTOLIC ARRAY
21
22 module systolic_array_2x2(
23     input clk,
24     input reset,
25     input [7:0] A1, A2,
26     input [7:0] B1, B2,
27     output [15:0] C1,
28     output [15:0] C2,
29     output [15:0] C3,
30     output [15:0] C4
31 );
32
33 wire [15:0] s1, s2, s3, s4;
34
35 // MAC1
36 mac_unit MAC1 (
37     .clk(clk),
38     .reset(reset),
39     .A(A1),
40     .B(B1),
41     .sum_in(s1),
42     .sum_out(s1)
43 );
44
45 // MAC2
46 mac_unit MAC2 (
47     .clk(clk),
48     .reset(reset),
49     .A(A2),
50     .B(B1),
51     .sum_in(s1),
52     .sum_out(s2)
53 );
54
55 // MAC3
56 mac_unit MAC3 (
57     .clk(clk),
58     .reset(reset),
59     .A(A1),
60     .B(B2),
61     .sum_in(s2),
62     .sum_out(s3)
63 );
64
65 // MAC4
66 mac_unit MAC4 (
67     .clk(clk),
68     .reset(reset),
69     .A(A2),
70     .B(B2),
71     .sum_in(s3),
72     .sum_out(s4)
73 );
74
75 C1 <= s1;
76 C2 <= s2;
77 C3 <= s3;
78 C4 <= s4;
79 endmodule
```

Simulation Results (Top Screenshot):

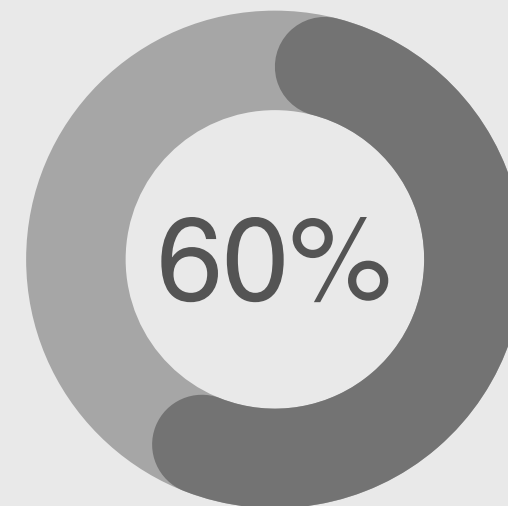
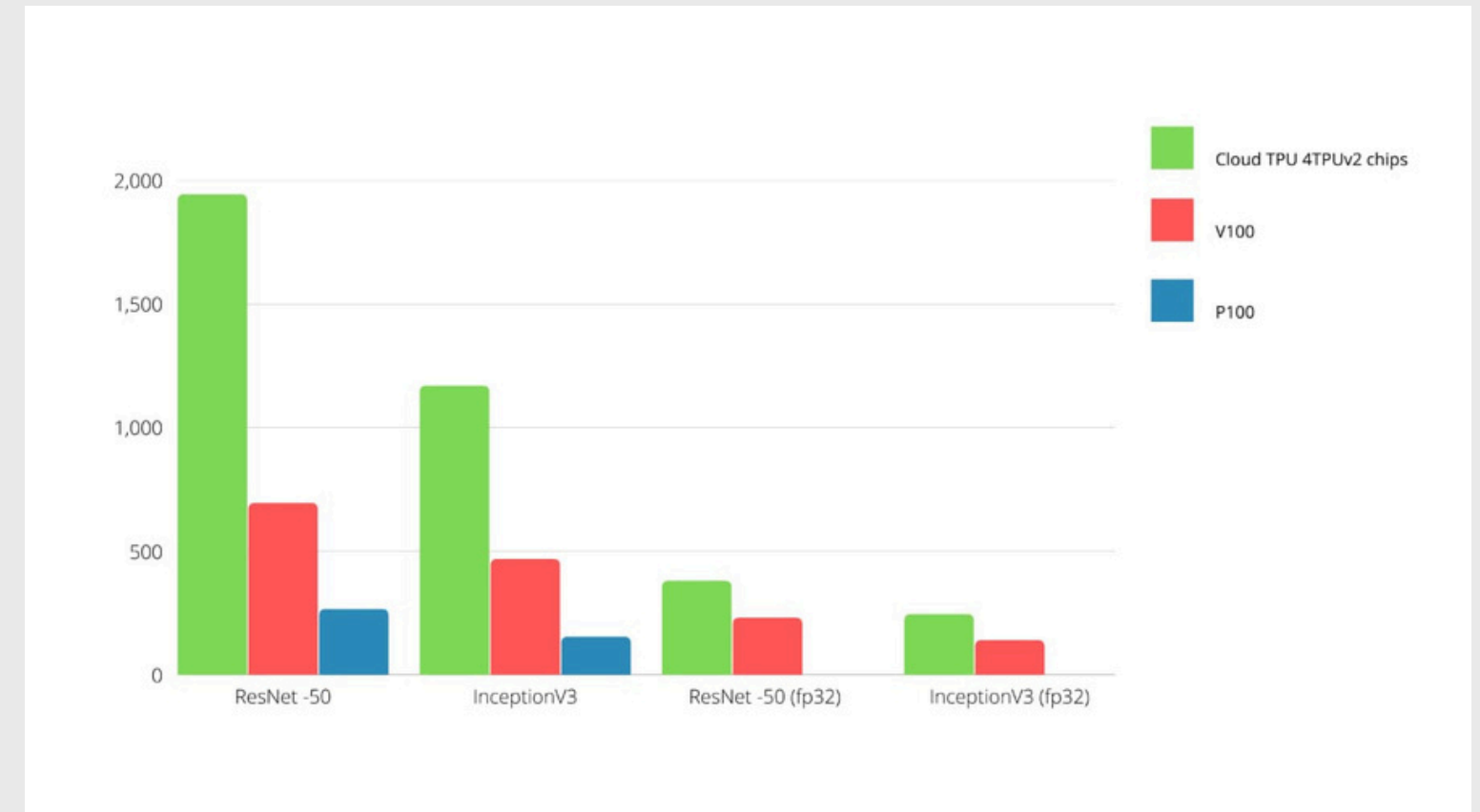
Time=0	C1=	C2=	C3=	C4=
Time=25	0	0	10	12

Simulation Results (Bottom Screenshot):

Time=0	C1=	C2=	C3=	C4=
Time=25	0	0	10	12

APPLICATIONS

- AI Accelerators
- Neural Network Processing
- Matrix Multiplication Operations
- Deep Learning Hardware
- Face Recognition Systems
- Image Processing Applications
- Speech Recognition Systems
- Edge AI Devices
- Autonomous Vehicles
- Google TPU Architectures



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PERFORMANCE**



THANK YOU

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